Robot Navigation Anomaly Detection

Team 2 | AI Algorithms | Mid-term Presentation | January 6, 2026

Agenda

- 1. Requirements & Changes (~1 min)
- 2. Metrics Overview (~2 min)
- 3. Interesting Insights from Anomaly Detection (~3 min)
- 4. Methodology for Causal Analysis & Results (~3 min)
- 5. Ideas for Future Improvements (~1 min)

1. Requirements & Changes

~1 min

Initial Approach (Problem)

- Log-based anomaly detection
- Dataset lacks RTF (Real-Time Factor) data
- Logs were not structured for analysis

Current Approach (Solution)

- CSV-based anomaly detection
- Structured, reliable data format
- Consistent across all runs

Data Sources

~1 min

- poses.csv — Ground truth and estimated poses
- behaviors.csv — Behavior tree execution logs
- rosbag2.csv — Raw ROS topics (AMCL pose, transforms, velocity commands)
- scenario.config — Scenario parameters

2. Performance Metrics

~2 min

Category	Metric	Unit	Description
Localization	mean_pos_error	m	Average position error
Localization	rmse_pos	m	Root mean square error (spike sensitive)
Localization	mean_yaw_error	rad	Average orientation error
Path	executed_path_length	m	Total path traveled
Path	path_efficiency	ratio	gt_path / executed (1 is best)
Kinematics	mean_linear_velocity	m/s	Average movement speed
Kinematics	trajectory_smoothness	rad/s2	Angular acceleration (lower = smoother)
Kinematics	duration	s	Total run time
AMCL	mean_amcl_uncertainty	m	Average localization confidence

Anomaly Detectors

~2 min

Rule-Based (8 types)

- goal_failure - Goal not reached
- no_initiation - Robot didn't start
- position_error_spike - Sudden errors
- stuck - Robot stopped moving
- high_amcl_uncertainty - Poor localization
- high_yaw_error - Orientation issues
- path_inefficiency - Sub-optimal path
- oscillation - Jerky motion (smoothness > 2.0)

ML-Based (1 type)

- Isolation Forest
- Captures multi-metric abnormal patterns that rules may miss.
- Uses all performance metrics

3. Interesting Insights from Anomaly Detection

~3 min

Anomaly Type	Count	% of Runs
goal_failure	150	50.0%
stuck	115	38.3%
Isolation Forest	45	15.0%
position_error_spike	15	5.0%
high_yaw_error	9	3.0%
path_inefficiency	7	2.3%

Key Findings

- Total occurrences: 341
- goal_failure is most common (50%)
- stuck follows at 38.3%
- Isolation Forest catches 15%
- Single run can have multiple anomalies

- Door-Width (120 Runs): High incidence of Goal Failure (64% of runs).
- Hallway-Window (120 Runs): Goal Failure remains the top anomaly (43% of runs).
- Room-Size (60 Runs): Agents frequently getting Stuck (52% of runs).

Insights - Anomaly Co-occurrence

~3 min

- position_error_spike + high_yaw_error (Jaccard = 0.60)
-   -> Poor orientation tracking leads to position drift
- high_yaw_error + path_inefficiency (Jaccard = 0.78)
-   -> Orientation errors cause inefficient path execution

Insights - Early Detection (First 30% of Run)

~3 min

Anomaly	PR-AUC	ROC-AUC	Cases	Insight
goal_failure	0.852	0.828	150	Highly predictable early
stuck	0.766	0.789	115	Highly predictable early
Isolation Forest	0.661	0.834	45	Good detection capability

Insights - Feature Signatures

~3 min

Goal Failure :

path_length : r=-0.491 ***
std_velocity : r=-0.437 ***
path_efficiency : r=-0.384 ***

Stuck :

duration : r=+0.524 ***
mean_angular_vel : r=-0.283 ***
path_length : r=+0.276 ***

high_amcl_uncertainty:

rmse_pos : r=+0.291 ***

max_velocity : r=+0.277 ***

std_pos_error : r=+0.274 ***

4. Causal Analysis - (Ensemble + Model Distillation / Surrogate Modeling)

- Step 1: High-Capacity Model Training (Ensemble)
- -> Voting Classifier: Decision Tree+ Random Forest + Gradient Boosting
- -> Prioritizes F1-score

- Step 2: Surrogate Model Training (Distilled Decision Tree)
- -> Constrained tree (max_depth=4)
- -> Target: Ensemble predictions (not ground truth)

Step 3: Rule Extraction

- -> Extract paths with P(Anomaly) > 0.5
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- Step 4: FOL Translation
- -> Map thresholds to semantic predicates

Derived FOL Rules - Examples

~3 min

Goal Failure:

(Cov=17.0%, Prec=1.00, Rec=0.34, n=51), Confusion: TP=51, FP=0, FN=99, TN=150

FOL: $\forall t$: in_corridor(t) $\land$ min_wall_distance(t) > 0.737 $\land$ obstacle_clearance_ratio(t) <= 17.970 $\Rightarrow$ goal_failure

NL: IF robot is in corridor AND min wall distance > 0.737 AND obstacle clearance ratio ≤ 17.970 THEN goal_failure

Stuck:

(Cov=6.7%, Prec=1.00, Rec=0.17, n=20), Confusion: TP=20, FP=0, FN=95, TN=185

FOL: $\forall t$: min_door_distance(t) > 1.792 $\land$ noise_level(t) <= 0.015 $\land$ min_door_distance(t) > 2.104 $\land$ room_area(t) <= 25.130 $\Rightarrow$ stuck

NL: IF min door distance > 1.792 AND noise level ≤ 0.015 AND min door distance > 2.104 AND room area ≤ 25.130 THEN stuck

Derived FOL Rules - Examples

High_amcl_uncertainty:

(Cov=4.0%, Prec=1.00, Rec=0.52, n=12), Confusion: TP=12, FP=0, FN=11, TN=277

FOL: $\forall t$: obstacle_clearance_ratio(t) > 31.314 $\wedge$ total_obstacle_area(t) <= 0.875 $\wedge$ door_width(t) > 0.935 $\Rightarrow$ high_amcl_uncertainty

NL: IF obstacle clearance ratio > 31.314 AND total obstacle area ≤ 0.875 AND door width > 0.935 THEN high_amcl_uncertainty

Path_inefficiency:

(Cov=3.3%, Prec=0.70, Rec=1.00, n=10), Confusion: TP=7, FP=3, FN=0, TN=290

FOL: $\forall t$: obstacle_clearance_ratio(t) <= 8.833 $\wedge$ clearance_ratio(t) > 6.812 $\wedge$ min_door_distance(t) <= 1.349 $\Rightarrow$ path_inefficiency

NL: IF obstacle clearance ratio ≤ 8.833 AND clearance ratio > 6.812 AND min door distance ≤ 1.349 THEN path_inefficiency

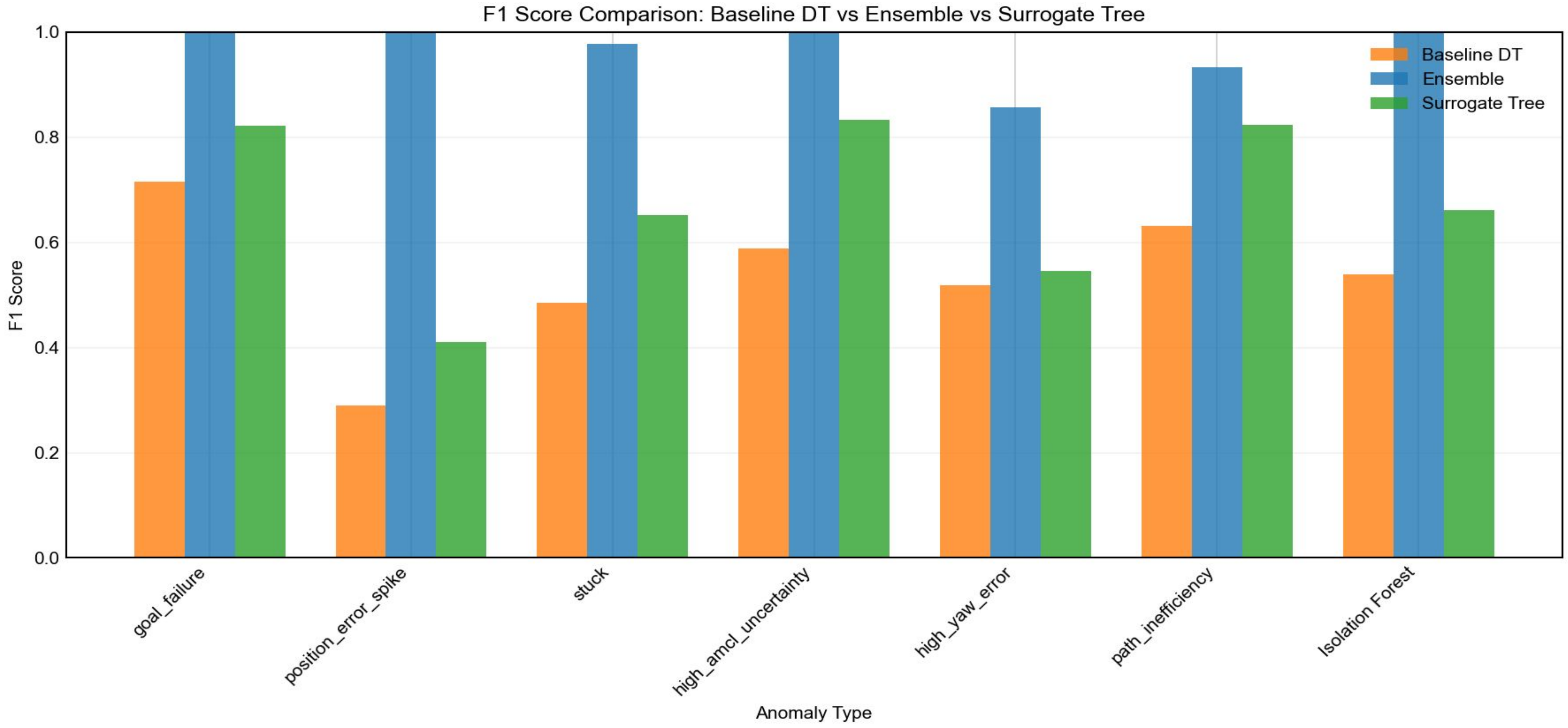
Derived FOL Rules - Examples

Isolation Forest:

(Cov=4.0%, Prec=1.00, Rec=0.27, n=12), Confusion: TP=12, FP=0, FN=33, TN=255

FOL: ∀t : obstacle_clearance_ratio(t) > 31.314 ∧ total_obstacle_area(t) <= 0.875 ∧ door_width(t) > 0.935 ⇒ Isolation Forest

NL: IF obstacle clearance ratio > 31.314 AND total obstacle area ≤ 0.875 AND door width > 0.935 THEN Isolation Forest



F1 Score Comparison: Baseline DT vs Ensemble vs Surrogate Tree
F1 Score
Anomaly Type
Baseline DT
Ensemble
Surrogate Tree
0.0
0.2
0.4
0.6
0.8
1.0
goal_failure
position_error_spike
stuck
high_amcl_uncertainty
high_yaw_error
path_inefficiency
Isolation Forest

5. Ideas for Future Improvements

~1 min

Short-term

- Hyperparameter tuning on surrogate tree
- Optimize max_depth, min_samples_leaf
- Better balance: simplicity vs accuracy

Long-term

- Feature engineering enhancements
- -> Add more obstacle-related features
- -> Include temporal dynamics
- Cross-validation on unseen scenarios
- Sensitivity analysis of rule thresholds

Thank You!